

PS-GAN: Pseudo Supervised Generative Adversarial Network With Single Scale Retinex Embedding for Infrared and Visible Image Fusion

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Abstract—Currently, ground truth fusion image, fused image contrast, and naturalness are rarely considered in existing infrared and visible image fusion (IVF) methods. In this article, we proposed a pseudosupervised generative adversarial network (GAN) with single scale retinex (SSR) embedding for IVF. First, a pseudoground truth fusion image conception and its computation method was proposed to solve ground truth fusion image shortage problem. Second, a novel SSR module embedded residual GAN was designed to improve fusion image contrast and naturalness. Finally, a special dense and mixed modal inputting strategy was also proposed for better modal mixed feature extraction. Extensive experimental results on public IVF datasets verified the superior performance of our proposed approach over other representative methods. It was demonstrated that the fused image details, contrast, and naturalness were significantly improved.

Index Terms—Generative adversarial network (GAN), image contrast, image naturalness, infrared (IR) and visible (VIS) image fusion (IVF), pseudoground truth fusion image (PGT).

I. INTRODUCTION

HYPERSPECTRAL or multimodal remote sensing images provide valuable resources for geoscience research [1], such as hyperspectral image superresolution [2], small infrared (IR) object detection [3], hyperspectral–multispectral image segmentation [4], and classification [5], [6]. Due to the availability of large number of remote sensing multimodal images, one of the emerging research problem is to fuse multimodal images from different modalities or sensors [1], such as infrared and visible image fusion (IVF).

In the last decade, IVF approaches have been proposed. They can be mainly categorized into two classes based on feature extraction techniques [7] as follows. 1) Handcrafted feature-based approaches (traditional methods). 2) Feature learning-based techniques (modern deep learning approaches).

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Our source codes are accessible at: <https://github.com/jinqijinqi/PSCM-GAN>
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Traditional IVF methods often use linear or nonlinear transformation [8], filtering process [9], image decomposition [10], or sparse representation [11] to extract image features and use simple maximizing, minimizing, averaging, or weighted summing to fuse image features. Although traditional IVF methods are simple and efficient, their performance is limited because no ground truth fusion information is used to guide the feature extracting and fusing processes.

On the contrary, modern deep learning IVF methods usually use source input images as pseudotarget images to guide feature extracting and fusing process due to the unavailability of ground truth fusion images [7]. These deep learning methods tried a variety of models, such as convolutional neural network (CNN) [12], attention mechanisms [13], autoencoders (AEs) [14], generative adversarial networks (GANs) [15], and transformer [16].

CNN-based IVF methods [12], [17], [18], [19], [20], [21] usually take infrared (IR) and visible (VIS) images as inputs and a fused image as an output. To take the advantage of machine learning power, the objective loss function is often constructed by comparing the fused image with the source input images [7]. Thus, the loss function plays an important role in CNN-based IVF methods and a lot of loss functions are built based on intensity similarity, structure similarity, gradient similarity, etc. The core part is to make sure that the fused image is not only similar with IR input image, but also with VIS input image. It leads to a similarity balancing problem between IR and VIS images. However, it remains an open question how to make a fused image that resembles both IR and VIS images [7].

Attention mechanism-based methods [13], [22], [23], [24] usually use spatial or channel attention to focus on salient regions during feature extraction or image reconstruction process. The attention mechanism has enormous potential to improve deep neural network-based IVF method performance.

Autoencoder (AE) is also used for IVF. Normally, IR and VIS images are inputted to an encoder for feature or latent feature extraction. Then, a decoder is used to reconstruct the input images [14], [25], [26], [27], [28]. Usually, manual fusion policy is used to fuse the encoded features and use the decoder to reconstruct the fused image [7], which is not fully machine learning process. Furthermore, there still exists the similarity balancing problem between IR and VIS images in decoder part.

On the contrary, GAN-based IVF methods [15], [29], [30], [31] use a generator to generate a fused image and a discriminator to make the fused image similar with the input IR and VIS

images. Compared to CNN and AE models, GAN has more flexibility to generate a fused image due to the less constrained adversarial loss usage. Usually, one discriminator is used to make sure that the generated image is similar to the input VIS image [31]. To avoid the biasing problem in one discriminator GAN, some methods tried two discriminators [15], [29], [30]. Two, however, there still exists the similarity balancing problem between IR and VIS images in both generator and discriminator parts.

Neural network transformer architecture has the potential to explore global relationship among data or data elements. Transformer-based IVF methods [16], [32], [33], [34], [35], [36], [37] usually extend the existing models by transformer architecture or build a fully transformer network [7].

Deep learning-based IVF methods achieved promising fusion results [7]. However, the two pseudotarget images (IR and VIS input source images) in deep learning methods not only lead to multiple target balancing problem, multiple loss function balancing problem, multiple discriminator balancing problem, but also lead to limited fusion image quality since the pseudotarget images are often low quality [7]. Some authors combined deep learning with image enhancement [38], [39] to improve fused image quality.

To address these problems, we proposed a pseudoground truth fusion image (PGI) conception and its computation method. A novel single scale retinex (SSR) embedded GAN was also proposed to efficiently improve image contrast and naturalness. Furthermore, we designed a special dense and mixed modal inputting strategy.

The main contributions of this study are as follows.

- 1) A novel concept of ground truth fusion, called PGI, and a PGI computation method were proposed.
- 2) A novel SSR embedded GAN, was designed.
- 3) The superior performance of our proposed IVF method was extensively validated using public datasets, with source codes made available for the research community.

The rest of this article is organized as follows. In Section II, the related work and motivation are presented. The proposed method is described in Section III. The experimental results and discussion are shown in Section IV. In Section V, the discussion is presented. Finally, Section VI concludes this article.

II. RELATED WORK AND MOTIVATION

GAN-based IVF methods have become one of the most important class of IVF approaches [15], [29], [30], [31]. Usually, a generator is used to generate a fused image and a discriminator is used to distinguish the generated image with the source IR and VIS images. Therefore, most of GAN-based IVF methods are unsupervised methods. The training is usually accomplished by an objective loss function that measure the similarity between the fused image and the source images [7].

The loss function in GAN-based IVF methods normally consists of “content loss” and “adversarial loss” components that measure the similarity from different perspectives [7]. “Content loss” component directly measures the similarity between fused image and source images. “Adversarial loss” component

indirectly reflexes the similarity between fused image and source images. Different from other models, such as CNN and AE models, GAN is more flexible to generate a fused image due to the usage of the less constrained adversarial loss.

FusionGAN [40] is the first GAN-based IVF method. One generator is used to generate a fused image and one discriminator is used to distinguish the fused image from source VIS image. Many extensions of the FusionGAN method are proposed by using attention mechanism, residual connections, different loss measurements, and multiple modal branches [7]. This one discriminator-based methods usually have a biasing problem, which forces the fused image to be similar to either VIS source image or IR source image.

To solve the biasing issue in single discriminator, GAN-based methods are extended to use two or more discriminators [15], [29], [30]. Normally, one discriminator forces the fused image to be similar to VIS source image and another discriminator forces the fused image to be similar to IR source image. The main advantage of using more discriminators is to preserve features in both source images. However, the balancing problem arises among multiple discriminators [7].

To deal with the shortage of ground truth fusion image in IVF task, most deep learning approaches simply took both IR and VIS input source images as two separate pseudotarget images to help deep model to learn automatic feature extraction and image reconstruction [7], [12]. It is well known that target image quality has a major impact on the performance of deep learning [7]. Apparently, the two pseudotarget images in most deep learning IVF methods are often low quality images and are much different from ideal ground truth fusion images. Therefore, the quality of fused images is limited, including salient object missing, salient region blur, salient detail loss, poor contrast, and poor naturalness.

To preserve salient or important region in fused image, some approaches use spatial attention or channel attention or transformer attention to weight feature maps [7], [13]. However, the salient foreground region may not be completely reserved in weighted feature maps during learning process without explicit salient region guiding information.

To preserve more details in fusion image, another line of literature uses image decomposition to explicitly obtain detail information [7], [41] or dense network architecture to allow detail information to flow from inputting ending to final outputting ending [12]. Although image decomposition is able to obtain image details, it does not provide solution for how to fuse the obtained detail information. Dense connection technique provides the possible path for detail information to propagate, but it does not guarantee that the image details will propagate to the output ending of the dense network.

To enhance fused image and improve its contrast and naturalness, some approaches use a postprocessing image enhancement step to improve fused image contrast and naturalness [39], [42]. However, the post processing deviates from the end-to-end training mechanism.

To extract modal features, some researchers design a network with two separate paths to extract IR and VIS features separately [15], [29]. Some other researchers design path crossing

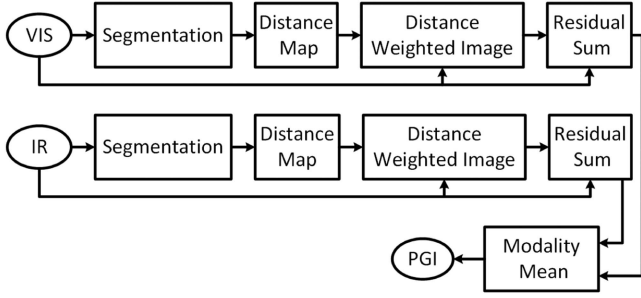


Fig. 1. Schematic diagram of our PGI computation process.

network to extract mixed modal features [30], [31]. It is shown in [30] and [31] that modal mixed features are better than modal separated features. However, path crossing network is more complicated than modal path separated network.

We observe that one of the biggest challenging problems in IVF is the shortage of ground truth fusion images. In this article, we tried to solve this problem by building a PGI. Specifically, we proposed a PGI conception and its computation method. Our IVF method uses our proposed PGI rather than input source images to guide our model training. The proposed PGI computation method guarantees that the salient objects and regions from both IR and VIS input images are emphasized and that detail information is also preserved in our PGI. To improve fused image contrast and naturalness, we proposed an SSR embedded GAN network. For modal mixed feature extraction, we proposed an efficient dense and mixed modal inputting strategy while keeping simple modal path separated network structure. Experimental results show that our proposed method is superior to other typical methods.

III. PROPOSED METHOD

In this section, we first introduce our PGI computation method, followed by the construction of our proposed pseudo supervised generative adversarial network (PS-GAN).

A. PGI Computation Method

The diagram of the whole process to compute PGI is shown in Fig. 1. The whole process contains five steps: segmentation, distance map generation, distance weighted image generation, residual sum, and modality mean. First, we normalize the input image intensity to range $[0, 1]$ and adopt a simple threshold method “Otsu” [43] to segment salient foreground region from input image I_{in} to obtain binary segmented image I_{sb} as shown in “segmented IR” and “segmented VIS” subimages in Fig. 2. Then, a distance map I_{dm} is computed from the segmented binary image I_{sb} as following:

$$I_{dm}(i, j) = \min_{m, n} \{ \sqrt{(i - m)^2 + (j - n)^2} : I_{sb}(m, n) = 0 \} \quad (1)$$

where $I_{sb}(m, n) = 0$ means the pixel (m, n) in image I_{sb} is background pixel and $I_{dm}(i, j)$ is the closest distance from pixel (i, j) to any background pixel (m, n) in the binary segmented image I_{sb} . As shown in the “distance IR” and “distance VIS” subimages in Fig. 2, it can be seen that the center of each salient

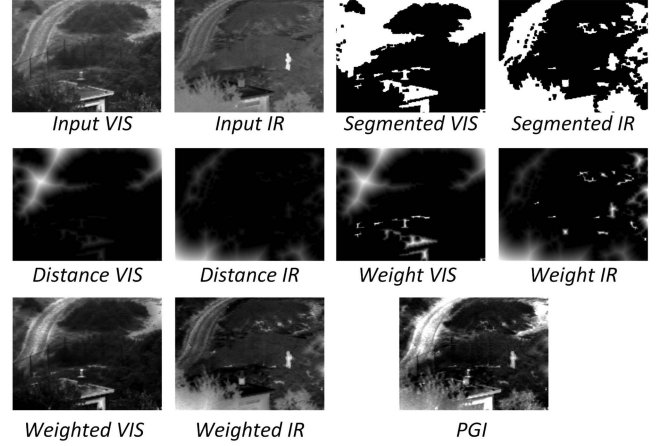


Fig. 2. Example results from PGI computation process. First row (from left to right): VIS source image, IR source image, segmented VIS image, and segmented IR image; Second row: distance VIS image, distance IR image, weight VIS image, and weight IR image; Third row: weighted VIS image, weighted IR image, and PGI image.

object takes the largest distance value and that salient object size has an important impact on the distance value of the object center. To deal with the salient object size problem, we need to normalize the distance map I_{dm} to obtain an object size free distance weight image I_{dw} in the following.

We assume that $C_{(i,j)}$ is the connected component containing pixel (i, j) in the binary segmented image I_{sb} . Then, a normalized distance weight image I_{dw} is computed by the following formula:

$$I_{dw}(i, j) = \begin{cases} 0, & \text{if } I_{dm}(i, j) = 0 \\ \frac{I_{dm}(i, j)}{\max\{I_{dm}(m, n) : (m, n) \in C_{(i,j)}\}}, & \text{otherwise} \end{cases} \quad (2)$$

where $I_{dw}(i, j)$ is the weight value in the pixel (i, j) and takes value in the range $[0, 1]$. From this formula, It can be seen that the center of each salient object takes value “1” in the normalized distance weight image I_{dw} , as shown in the “weight IR” and “weight VIS” subimages in Fig. 2. It is apparent that our normalized distance weight image I_{dw} is salient object size free and the salient “person” in “weight VIS” subimages in Fig. 2 is emphasized.

A distance weighted image I_{dwi} of a source input image I_{in} is the pixelwise product of the source input image I_{in} and its normalized distance weight image I_{dw} as shown in the following:

$$I_{dwi} = I_{dw} \cdot I_{in}. \quad (3)$$

It is obvious that the distance weighted image I_{dwi} only contains salient objects since the distance weight image I_{dw} takes “0” value in background area, as shown in the “weight IR” and “weight VIS” subimages in Fig. 2. We have to recover the image information in background area in the following.

Inspired by the idea of residual connection in deep learning, we proposed a residual summation to recover image information in background area. Specifically, our residual summation image I_{rs} is computed as the residual summation of input image I_{in} as

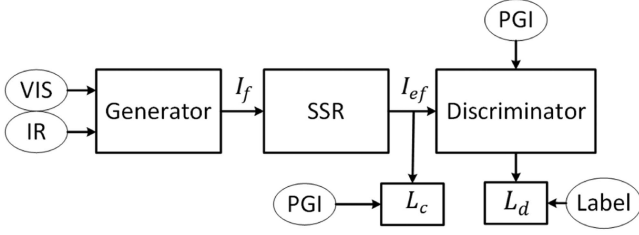


Fig. 3. Schematic diagram of our proposed PS-GAN: “IR” and “VIS” are input images; “ I_f ” and “ I_{ef} ” are fused image and enhanced fusion image, respectively; “ L_c ” and “ L_d ” computes the content and discrimination losses for PGI and enhanced fusion image I_{ef} , respectively; “Label” takes binary values (“1” for PGI, and “0” for generated image “ I_{ef} .”

follows:

$$I_{rs} = I_{dwi} + I_{in} = I_{dw} \cdot I_{in} + I_{in}. \quad (4)$$

From this formula, it is easy to see that our residual summation image I_{rs} not only emphasizes salient foreground area, but also preserves the original image information in background area, as shown in the “weighted VIS” and “weighted IR” subimages in Fig. 2

Finally, we denote IR residual summation image and VIS residual summation image by I_{rs}^{IR} and I_{rs}^{VIS} , respectively. Our PGI is defined as the following modality mean of I_{rs}^{IR} and I_{rs}^{VIS} :

$$PGI = \frac{I_{rs}^{IR} + I_{rs}^{VIS}}{2}. \quad (5)$$

Then, PGI intensity is normalized to range [0 1] by linear scaling. It can be seen that our PGI not only emphasizes the salient foreground regions, but also preserves the original detail information in background area from both VIS and IR modalities, as shown in the subimage “PGI” in Fig. 2.

Fig. 2 shows an example of all intermediate and final results for a pair of IR and VIS images during our PGI computation process. It is apparent that PGI image in the last row in Fig. 2 emphasizes both IR and VIS salient foreground regions in addition to the preservation of detail information in background regions from both IR and VIS source images. Therefore, we use PGI image as our ground truth fusion image to guide our network training and convert the usual unsupervised IVF task to a supervised IVF task.

B. Proposed PS-GAN

In this section, we first show the overall schematic diagram of our proposed PS-GAN structure in Fig. 3 followed by details of each component in the PS-GAN.

Our PS-GAN consists of three main parts: one generator, one SSR component, one discriminator as shown in Fig. 3. We embedded the SSR component between generator and discriminator. “ L_c ” and “ L_d ” in Fig. 3 represent two loss computation positions. “ L_c ” computes the content loss between PGI and fused image I_f . “ L_d ” computes the discrimination loss for PGI and enhanced fusion image I_{ef} .

1) *Generator Architecture*: We designed a dual-encoder-single-decoder generator, as shown in Fig. 4. The two encoders, E_{IR} , E_{VIS} , share the same structure with each other.

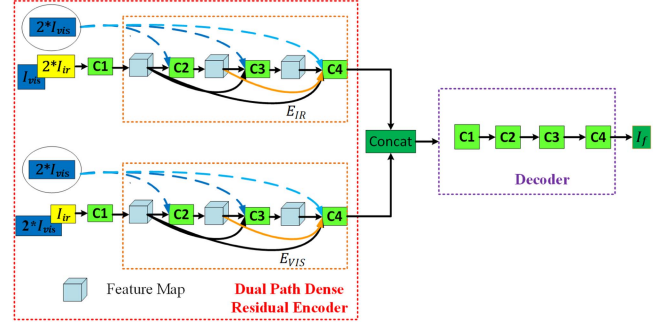


Fig. 4. Generator network architecture. “ $2 * I_{VIS}$ ” and “ $2 * I_{IR}$ ” represent two VIS source images and two IR source images concatenated in channel dimension, respectively. C_1, C_2, C_3 , and C_4 are convolution layers. For each dense layer, the multiple inputs indicated by multiple arrows in front of each layer are concatenated in the channel dimension.

However, we arrange different inputs to two encoders to encourage E_{IR} and E_{VIS} to focus more attention to IR and VIS images, respectively. Each encoder consists of four convolution layers: C_1, C_2, C_3 , and C_4 . The four convolution layers have 128, 256, 64, and 32 kernels, respectively. They have kernel size $5 \times 5, 5 \times 5, 3 \times 3$, and 3×3 , respectively. We use dense residual connections for all convolution layers to allow image detail information to flow to any layer in our generator. The spatial size across different layers is the same.

After the two encoders, one concatenation layer is used to concatenate the outputs of the two encoders as inputs to the decoder. The decoder consists of four convolution layers with 64, 32, 16, and 1 output channels, respectively. All convolution layers in the decoder use the same kernel size 3×3 . The spatial size across different layers is the same. The output of the decoder is our fused image.

The activation layer for each convolution layer in our proposed generator is leaky ReLU except the last convolution layer in the decoder followed by a Tanh layer.

2) *Dense and Mixed Modal Inputting Strategy*: To further avoid image detail information loss, our dense inputting strategy simply concatenates two VIS images in channel as inputs to each layer of C_2, C_3, C_4 , as shown in Fig. 4. Thus, rich detail information from VIS image can reach any convolution layer in encoder.

Our modal mixed inputting strategy simply concatenates two IR images and one VIS image in channel as inputs to encoder E_{IR} and concatenates two VIS images and one IR image in channel as inputs to the encoder E_{VIS} , as shown in Fig. 4. Thus, each of our two encoders E_{IR} and E_{VIS} is able to extract modal mixed features. However, encoder E_{IR} biases to IR modality and E_{VIS} biases to VIS modality.

3) *SSR Component*: As shown in Fig. 5, our SSR component contains Gaussian filtering(convolution), logarithm, and minus operations [44]. We embedded an SSR component into our GAN between the generator and the discriminator, as shown in Fig. 3. The output I_{ef} of SSR component is the enhanced image of fusion image I_{fused} as follows:

$$I_{ef} = \log I_{fused} - \log (F \otimes I_{fused}) \quad (6)$$

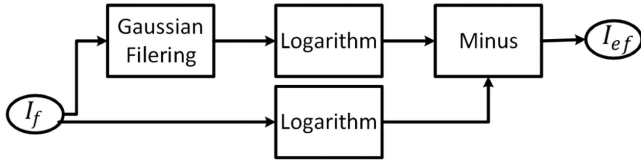


Fig. 5. SSR structure.

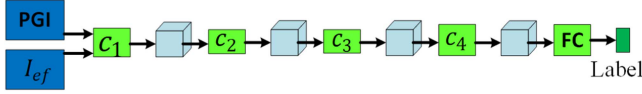


Fig. 6. Discriminator network architecture. “PGI” and “ I_{ef} ” are PGI and enhanced fusion image, respectively. C_1 , C_2 , C_3 , and C_4 are convolution layers. “FC” is fully convolution layer. “Label” takes binary values (“1” for PGI, and “0” for generated image “ I_{ef} ”).

where I_{ef} , $\otimes$, and F are the enhanced fusion image, convolution operator and Gaussian filter, respectively. The Gaussian filter F is defined as

$$F(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x^2 + y^2)}{\sigma^2}\right) \quad (7)$$

where (x, y) are pixel coordinates in the integer and σ^2 is set to 16 in this study.

The embedded SSR component is able to enhance the contrast [44] of our fused image I_f , as shown in Fig. 3.

4) *Discriminator Architecture*: As shown in Fig. 6, our discriminator network consists of four convolution layers and one fully connected layer. All four convolution layers have the same kernel size 3×3 . The number of kernels in the four convolution layers are 32, 64, 128, and 256, respectively. Each convolution layer is followed by a leaky ReLU layer and a max-pooling layer. The last fully connected layer is followed by a sigmoid layer. Our discriminator is to discriminate our PGI (real) from the enhanced fusion image I_{ef} (fake) from the SSR component.

Different from other multiple discriminator-based GAN method [15], our proposed method only uses one discriminator and avoids the balancing problem among multiple discriminators.

C. Loss Function

The loss function of our PS-GAN consists of both generator loss and discriminator loss functions. In discriminator, we discriminate our pseudoground truth image (PGI) from enhanced fusion image I_{ef} , as shown in Fig. 3. In generator, we define a loss to encourage the enhanced fusion image I_{ef} to be similar to our PGI.

1) *Discriminator Loss Function*: In order to distinguish the enhanced fusion image I_{ef} from our PGI, we use the least MSE to define the discriminator loss function L_D as follows:

$$L_D = \frac{1}{N} \sum_{n=1}^N (D(\text{PGI}^n) - b)^2 + \frac{1}{N} \sum_{n=1}^N (D(I_{ef}^n) - a)^2 \quad (8)$$

where $b = 1$, $a = 0$, N is the batch size during training. PGI^n represents the n th PGI. I_{ef}^n represents the n th enhanced fusion image. $D(\text{PGI}^n)$ and $D(I_{ef}^n)$ are the discriminator outputs for PGI^n and I_{ef}^n , respectively.

The optimal discriminator D^* is found by minimizing the discriminator function L_D as following:

$$D^* = \text{argmin}_D L_D. \quad (9)$$

2) *Generator Loss Function*: The generator loss L_G consists of two parts: class mean square loss L_{cms} and content loss $L_{content}$. It is defined as follows:

$$L_G = L_{cms} + \gamma \cdot L_{content} \quad (10)$$

where γ is a hyperparameter and empirically set to 100 in this article.

The class mean square loss L_{cms} is different from usual cross entropy (CE) loss and defined as follows:

$$L_{cms}(G) = \frac{1}{N} \sum_{n=1}^N (D(G(\text{VIS}^n, \text{IR}^n)) - c)^2 \quad (11)$$

where $c = 1$, VIS^n , IR^n are the n th VIS input image and the n th IR input image, respectively. $G(\text{VIS}^n, \text{IR}^n)$ is the output of generator G for the inputs VIS^n and IR^n . $D(G(\text{VIS}^n, \text{IR}^n))$ is the output of discriminator D for the input $G(\text{VIS}^n, \text{IR}^n)$.

In some existing image fusion methods, authors usually use mean square error (MSE) loss for content loss $L_{content}$ between fused image and the input IR and VIS images [7]. However, this loss function has the following shortcomings.

- 1) MSE loss is more sensitive to large errors, which will make the fused image tend to be smooth.
- 2) The network directly imposes pixel-level equation constraints between the fused image and the input images (IR and VIS images) without considering the image's overall structure, which will make the fused image tend to be an average image of IR and VIS input images.

On the contrary, in this article we discard the MSE loss function and define the content loss $L_{content}$ as the following weighted sum of structure similarity loss L_{ss} and gradient similarity loss $L_{gradient}$ between the enhanced fusion image and our PGI:

$$L_{content} = L_{ss} + \alpha L_{gradient} \quad (12)$$

where α is a hyperparameter and empirically set to $\alpha = 5$ in this article, and L_{ss} and $L_{gradient}$ are defined as following:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y)(2\delta_{xy})}{(\mu_x^2 + \mu_y^2)(\delta_x^2 + \delta_y^2)} \quad (13)$$

$$L_{ss} = 1 - \text{SSIM}(I_{ef}, \text{PGI}) \quad (14)$$

$$\text{Gradient}(x, y) = \frac{1}{M} \sum_{n=1}^M (\nabla_x - \nabla_y)^2 \quad (15)$$

$$L_{gradient} = \text{Gradient}(I_{ef}, \text{PGI}) \quad (16)$$

where μ , δ , M , and ∇ represents mean, standard deviation (SD) (or covariance), the number of pixels, and gradient calculation operation, respectively.

The optimal generator G^* is found by minimizing the generator loss function L_G as following:

$$G^* = \operatorname{argmin}_G L_G. \quad (17)$$

Most of existing IVF methods, which often use two inputting source images VIS and IR as two PGIs, define structure and gradient similarity loss functions between fused image and each of the two PGIs, which results into many content loss function components [7]. However, our proposed GAN-based method only define structure and gradient similarity loss functions between fused image and our unique PGI, which halves the number of content loss function components.

IV. EXPERIMENTAL RESULTS

In this section, experimental setup, ablation study, qualitative, and quantitative comparisons are detailed.

A. Experimental Setup

The proposed method in this article was evaluated both qualitatively and quantitatively in three public IVF benchmark datasets: the TNO [45], LLVIP [46], and VOT2020-RGBT [47]. We chose 11 metrics and other 16 representative methods for comparison. The 16 typical methods chosen for comparison cover a variety of categories include the following.

- 1) Classical image processing-based techniques that do not rely on deep learning, such as CBF [48], Structaware [49], MDlattLRR [50], and HybridMSD [51]. They use filter or image decomposition techniques to extract and fuse the image features.
- 2) Methods based on an AE for deep learning, such as DenseFuse [52], NestFuse [53], and RFN-Nest [54]. They employ an encoder for feature extraction and feature fusion at the latent layer.
- 3) Methods based on deep learning with a one-step fusion network, such as low rank guided CNN (LRG-CNN) [55], CNN [56], VGGML [57], PMGI [58], ResNet-Zca [59], and U2Fusion [60].
- 4) Methods based on GAN for deep learning, such as CrossFuse-GAN [61], DDcGAN [62], MEF-GAN [63]. The results from these 16 methods are obtained by running their open-source codes with recommended setting up in their corresponding studies

The chosen 11 metrics are edge intensity (EI), CE, spatial frequency (SF), entropy (EN), quality of images ($Q^{ab/f}$), SF sum of correlation coefficients (SCDs), noise in images ($N^{ab/f}$), mutual information between images (MIab/f), SD, definition (DF), and a novel visual information fidelity (VIFF) [7].

A larger EI and SF indicate more edges and detailed information in the fused images. A larger EN value indicates more information in the fused image. Larger SCD and MIab/f indicate that the fused image retains sufficient information from the source images and is more similar to the source images. A larger VIFF value indicates that the fused image is more natural or realistic. The large DF and SD values indicate that the contrast of the fused image is high. A large $Q^{ab/f}$ value indicates good image quality. Large $N^{ab/f}$ means a noisy image. A large CE

value indicates a significant difference between the source and fused images. In summary, large metrics values, except for CE and $N^{ab/f}$, mean better fused images [7].

We randomly selected 59 pairs of different IR images and VIS images from the TNO, LLVIP, and VOT2020-RGBT database for training and validation. The image spatial size in TNO dataset is not the same and it varies from 256×256 to 678×917 . To augment the training dataset, we cropped each source image with a 120×120 pixel window and a 14 pixel stride to obtain all possible sub-images in all possible positions. Finally, we obtained 54 842 pairs of subimages, 70% of which are used for training and the remaining for validation.

We used Linux Ubuntu OS, Intel(R) Xeon(R) CPU E5-2687W v3 @ 3.10GHz, and Nvidia GeForce RTX 2080 TI GPU for training and testing. The algorithm was implemented based on the PyTorch framework. The network was trained with a batch size $B = 32$ and Adam optimization with a learning rate of 0.0002, the first exponential decay of 0.9, the second exponential decay of 0.999, and a small positive number of $1e - 8$ for preventing division by zero. The SSR does not has any learning parameters and the automatic differentiation mechanism in Pytorch framework was used when computing gradient of loss related to the SSR component.

For computation complexity, we measured the time for testing. A set of 21 pairs of IR and VIS images with resolution 768×576 were used to test the time performance of our trained generator network. It took about 10 s to generate the 21 fused images. The average testing time for each pair of IR and VIS images is about 0.5 s.

For each training iteration, we trained the discriminator twice and the generator once. The training took approximately 24 h with 30 epochs. In the test phase, the discriminator was removed and the generator and the SSR component were used to obtain the enhanced fused images of testing images. We directly inputted the original test images to the trained generator and SSR component without any cropping since the generator network is a fully convolutional network.

B. Ablation Study

To demonstrate the effectiveness of the proposed different generator loss function components and SSR component in our method, five ablation experiments, consisting of some combinations of SSR and generator loss function components, were performed.

- 1) "Abl. 1" (effectiveness of L_{cms}): $\{L_G = L_{cms}\}$.
- 2) "Abl.2" (effectiveness of $L_{gradient}$): $\{L_G = L_{cms} + \gamma * L_{gradient}\}$.
- 3) "Abl.3" (effectiveness of L_{ss}): $\{L_G = L_{cms} + \gamma * (L_{ss})\}$.
- 4) "Abl. 4" (effectiveness of combination of $L_{ss}, L_{gradient}$): $\{L_G = L_{cms} + \gamma * (L_{gradient} + \alpha * L_{ss})\}$.
- 5) "Abl. 5" (effectiveness of SSR): $\{SSR, L_G = L_{cms} + \gamma * (L_{gradient} + \alpha * L_{ss})\}$. The discriminator loss L_D is the same in five ablation studies.

We comparatively show some exemplar results from the five ablation experiments "Abl. 1," "Abl. 2," "Abl. 3," "Abl. 4," and "Abl. 5" in the third, fourth, fifth, sixth, and seventh columns in



Fig. 7. Ablation study exemplar results. From left column to right column: IR input images, VIS input images, Abl. 1 fused images, Abl. 2 fused images, Abl. 3 fused images, Abl. 4 fused images, and Abl. 5 fused images.

Fig. 7, respectively. The first and second columns in Fig. 7 are the IR and VIS input images, respectively. It can be observed from Fig. 7 that the fused images in the third column from “Abl. 1” using only class MSE loss L_{cms} are dark and some details are lost (such as trees, buildings, and grasses). It is very common that generated image from GAN is blurry with only adversarial loss. The fused images in the fourth column in Fig. 7 from “Abl. 2” recover details lost in “Abl. 1.” It demonstrates the effectiveness of gradient similarity loss $L_{gradient}$. The fused images from “Abl. 3” in the fifth column in Fig. 7 show more details than the images from “Abl. 2.” It is illustrated that the structure similarity loss L_{ss} is more effective than gradient similarity loss $L_{gradient}$. Fused images from “Abl. 4” in the sixth column in Fig. 7 preserve not only image details but also the salient foreground regions (such as IR person and helicopter). It demonstrates that the combination of gradient similarity loss $L_{gradient}$ and structure similarity loss L_{ss} is more effective than the single one of them. In addition to preserving image details and image salient foreground regions from both IR and VIS input images, the fused images from “Abl. 5” in the seventh column in Fig. 7 have much better contrast, naturalness than those fused ones in other ablation studies. It demonstrates the effectiveness of SSR for image contrast and naturalness improvement. These observations are also consistent with our quantitative results in the following section.

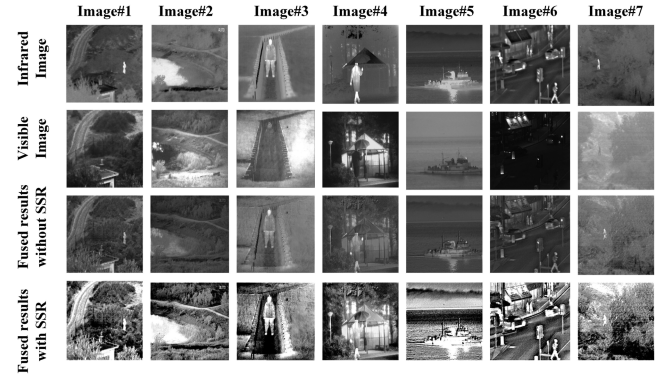


Fig. 8. Example results in TNO dataset: from top row to bottom row: input IR images, input VIS images, fused results without SSR, and fused results with SSR.

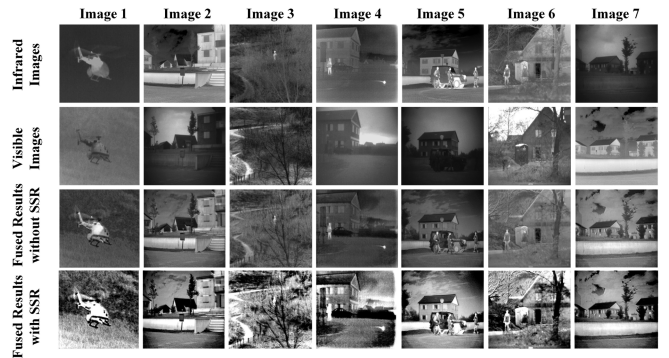


Fig. 9. Example results in VOT2020-RGBT dataset: from top row to bottom row: input IR images, input VIS images, fused results without SSR, and fused results with SSR.

In summary, it is demonstrated from the ablation study results in Fig. 7 that structure similarity loss L_{ss} has better capability to preserve details than gradient loss $L_{gradient}$ and that “Abl. 4” using the three generator losses (class MSE loss L_{cmse} , structure similarity loss L_{ss} , and gradient loss $L_{gradient}$) preserves both image details and salient foreground regions well. It can be also seen from Fig. 7 that our proposed method (“Abl. 5”) achieves best results in terms of image contrast and content preservation. Therefore, we used SSR and all three generator losses (class MSE loss L_{cmse} , structure similarity loss L_{ss} , and gradient loss $L_{gradient}$) in our proposed method in this article.

To further demonstrate the effectiveness of SSR component, we show some exemplar results from our proposed method with and without SSR component over three public datasets TNO [45], VOT2020-RGBT [47], and LLVIP [46] in Figs. 8–10, respectively. The IR images, VIS images, output fused images without SSR component (i.e., I_f in Fig. 3), and output fused images with SSR component (i.e., I_{ef} in Fig. 3) are shown in the first row, the second row, the third row, and the fourth row in Figs. 8–10, respectively. It can be seen from Figs. 8–10 that the fused images without SSR component in the third row have good capability to preserve details and salient foreground regions (such as persons, helicopters, and cars). However, the fused images with SSR component in the fourth row not only have the same capability to preserve details and salient foreground

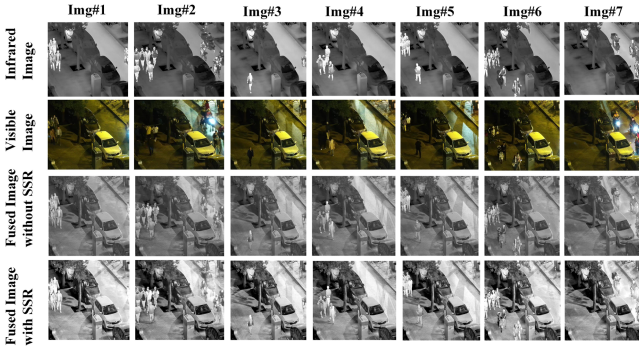


Fig. 10. Example results in LLVIP dataset: from top row to bottom row: input IR images, input VIS images, fused results without SSR, and fused results with SSR.

TABLE I
QUANTITATIVE RESULTS OF FIVE ABLATION STUDIES IN THE DATASET
TNO [45]

Ablation Studies	EF $\uparrow$	CE $\downarrow$	SP $\uparrow$	EN $\uparrow$	Qab/f $\uparrow$	SCD $\uparrow$	Nab/f $\downarrow$	MIab/f $\uparrow$	SD $\uparrow$	DF $\uparrow$	VIF $\uparrow$
Abl. 1	51.22	1.36	11.68	7.14	0.11	0.23	0.15	14.28	70.54	4.96	0.06
Abl. 2	58.52	0.74	14.55	6.65	0.11	0.35	0.17	13.30	72.05	6.05	0.11
Abl. 3	78.76	0.40	18.03	6.87	0.12	0.35	0.25	13.75	63.55	8.99	0.13
Abl. 4	79.84	0.36	18.14	6.78	0.12	0.33	0.25	13.57	63.75	9.06	0.13
Abl. 5	92.80	1.23	21.21	7.74	0.25	1.40	0.32	15.48	68.75	10.73	1.22

regions as the fused images without SSR component in the third row, but also have better image quality (such as richer details, higher contrast, and naturalness) which may be helpful in downstream tasks, such as object detection and classification. It is illustrated from Figs. 8–10 that our SSR component plays an important role in improving fused image quality and that our proposed method achieves good experimental results over three public datasets, which are also observed from our quantitative experimental results in the following section.

To quantitatively show the effectiveness of our loss function components and SSR component, we also computed the values of the chosen 11 metrics from our five ablation studies in the TNO dataset [45], which are shown in Table I. It can be seen from the metrics in the Table I that the overall performance is increasing from Abl. 1 to Abl. 5. It is demonstrated quantitatively that the structure similarity loss component is more effective than gradient similarity loss component and that the combination of them is more effective than single one of them, and that Abl. 5, using all loss components and SSR component, performed best. These observations are consistent with the observations from the exemplar results in Figs. 8–10.

Based on these ablation experiments, it can be observed that all three generator loss functions and the SSR component play a very important role in the fusion process, and that our proposed approach, consisting of the three generator loss functions and the SSR component (in ablation 5), performs the best among the five ablation studies.

C. Comparison With Other Methods: Visual Examples

To visually compare our method with other typical methods, we present the comparative visual example results of our method with other 16 methods in Figs. 11–14.

In Fig. 11, we observe that all these methods can preserve the salient “man” (shown in the green box) in the salient foreground

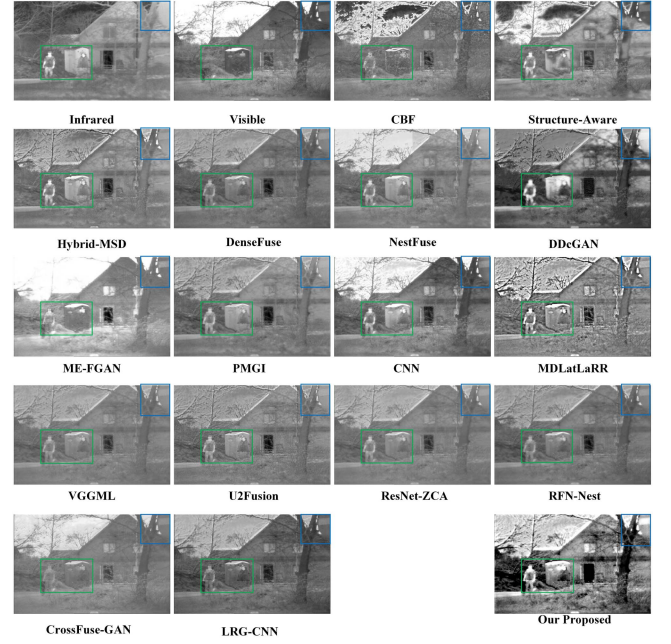


Fig. 11. Comparison results with other representative methods: (from top to bottom and left to right) input IR images, input VIS images, fused results from the 16 typical methods, and our fused result.

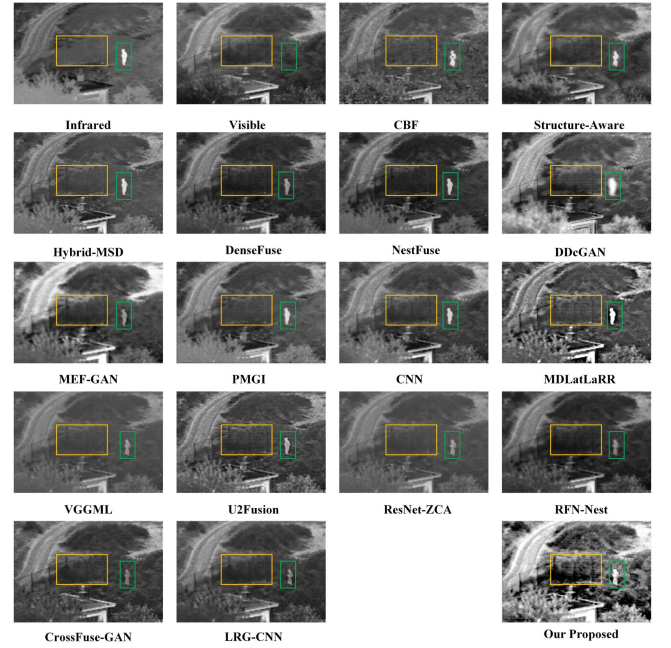


Fig. 12. Comparison results with other typical methods: (from top to bottom and left to right) input IR images, input VIS images, fused results from the 16 typical methods, and our fused result.

region from the IR image in the final fused images. However, our preserved “man” has better contrast than others, which is helpful for downstream object detection task (such as person detection). Furthermore, the salient foreground region (such as sky) from VIS image is better preserved in our method than most of others, as shown in the region in the blue square. It can be seen from Fig. 11 that both our proposed method and the

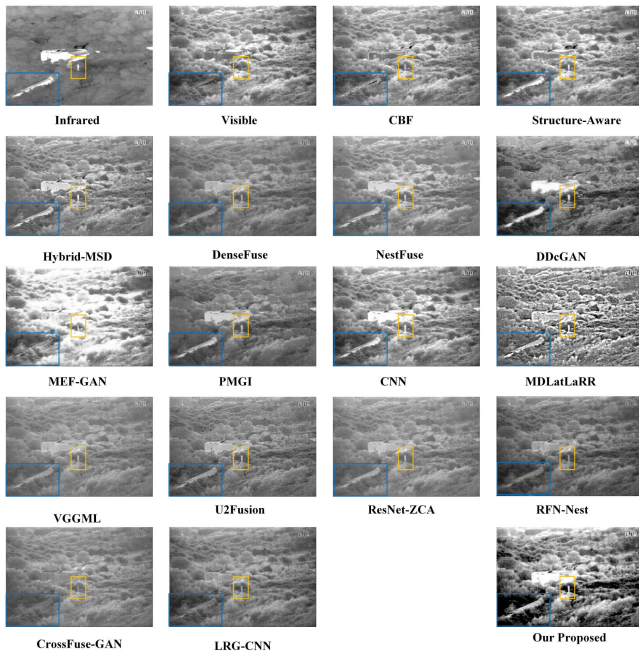


Fig. 13. Comparison results with other representative methods: (from top to bottom and left to right) input IR images, input VIS images, fused results from the 16 typical methods, and our fused result.



Fig. 14. Comparison results with other representative methods: (from top to bottom and left to right) input IR images, input VIS images, fused results from the 16 typical methods, and our fused result.

CNN method [56] preserves both salient IR and VIS foreground regions very well. However, the overall contrast of the fused image from our method is better than the one from the CNN method [56]. In addition, the salient objects (such as person, house, window, door, and tree) in our fused image are more apparent than in others and our fused image is more natural and realistic in terms of visualization.

According to Fig. 12, our fused image is better than the fused results from the other methods in terms of appearance and detailed information preservation. The overall contrast of our fused image is higher than the contrast of other fused images from other methods. The salient foreground regions from both IR image (such as “person”) and VIS image (such as road and building) are well preserved in our fused image. Especially, image details such as the fence and its poles from VIS image are better preserved in our method than in other methods, as shown in the yellow rectangle. It can be seen from Fig. 12 that both salient objects and details are preserved very well in our fused image and that our fused image is more visually pleasant than others.

As shown in Fig. 13, our fused image has better appearance and preserves more detailed information than other results from other methods. The overall contrast of our fused image is higher than the contrast of other fused images from other methods. The salient foreground regions from both IR image (such as bunker, road, “person”) and VIS image (such as the trees) are well preserved in our fused image, as shown in the blue or yellow rectangles. Furthermore, the contrast between the preserved IR foreground region and VIS foreground region is well automatically adjusted in our fused image and it leads to better visual pleasant of our fusion image than others. Except the preserved salient foreground objects, many image details such as trees from VIS image are better preserved in our method than in other methods. Therefore, our proposed method has a strong ability to preserve salient objects and details and improve the overall appearance of the fused images. Fig. 14 shows that our fused image looks better than other fused images in terms of appearance and detailed information preservation. The salient foreground regions from both the IR image (such as “person”) and VIS image (such as the “garden lamp” and the lightning houses) are well preserved in our fused image. The tree in the blue box deteriorates in the fused results from all other methods and is accurately preserved in our method. The “garden lamp” in the yellow box and the “bench” in the green box are preserved in all methods. The “plants” on the bottom left have apparent artifacts in the results from other methods. Our proposed method can accurately preserve the “plants” on the bottom left. Compared to other results, our fused image has better visual pleasant. Therefore, our proposed method has a strong ability to preserve salient foreground regions, objects, and details and improve the overall appearance of the fused images.

From Fig. 15, it can still be observed that our fused image is better than the fused results from the other methods in terms of appearance and detailed information. The salient foreground regions from both the IR image (such as “persons,” “cars,” “street lamps,” and “traffic lights”) and the VIS image (such as “store lights,” “signboard,” and “letters over signboard”) are well preserved in our fused image. The “person with backpack” in the green box is preserved in all methods, including ours. The “table and stand chairs” in the yellow boxes can be seen in our fused image and are difficult to see in the fused images from all other methods. The “house” in the blue box is well preserved in the fused image. However, the details of the “house” are lost in other fused images from other methods. Some artifacts were introduced to the “house” in the blue boxes in other fused images

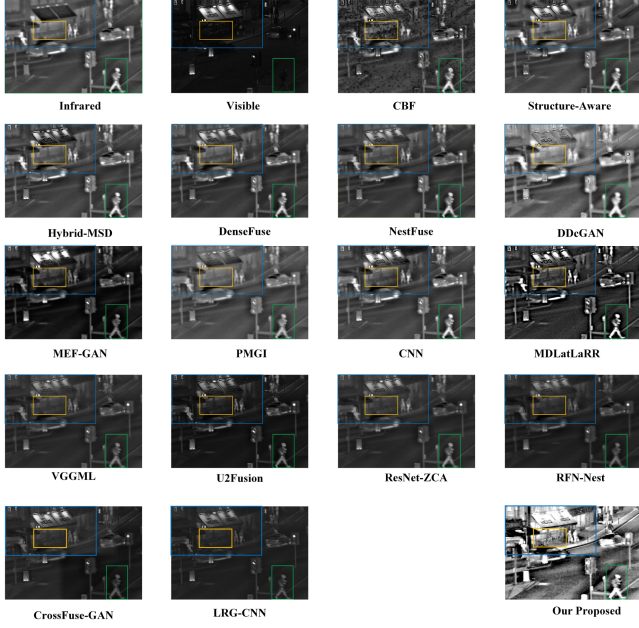


Fig. 15. Comparison results with the state-of-the-art methods: (from top to bottom and left to right) input IR images, input VIS images, fused results from the 14 state-of-the-art methods, and our fused result.

TABLE II
QUANTITATIVE COMPARATIVE RESULTS IN TNO DATASETS: THE BEST, SECOND-BEST, AND THIRD-BEST VALUES IN THE TABLE ARE HIGHLIGHTED IN RED **BOLD**, GREEN *ITALICS*, AND UNDERLINED BLUE, RESPECTIVELY

Methods	Year	EI†	EN†	MIab/f†	SD†	DF†	VIFF†
CBF [48]	2015	52.30	1.32	<u>11.09</u>	6.83	0.45	1.32
StructAware [49]	2018	44.91	<i>0.92</i>	11.20	7.05	0.62	1.26
HybridMSD [51]	2016	46.32	1.32	12.35	6.71	<i>0.50</i>	1.58
DenseFuse [52]	2018	36.48	1.40	9.32	6.85	0.47	1.53
NestFuse [53]	2020	38.44	1.45	9.71	6.89	0.50	1.58
DDxGAN [62]	2020	<u>54.82</u>	1.10	12.84	<i>7.42</i>	0.36	1.38
MEF-GAN [63]	2021	36.91	1.50	7.85	6.97	0.21	1.31
PMGI [58]	2020	37.21	1.57	8.72	6.87	0.38	1.57
CNN [56]	2017	44.83	0.89	<i>11.65</i>	7.06	<i>0.58</i>	1.60
MDLatLRR [50]	2020	<i>46.89</i>	<i>1.02</i>	21.62	<i>7.02</i>	0.32	1.56
VGGML [57]	2018	24.01	1.53	6.41	6.18	0.33	<i>1.60</i>
U2Fusion [60]	2020	48.49	1.33	11.04	6.72	0.39	<i>1.59</i>
ResNet-ZCA [59]	2019	26.24	1.62	5.92	6.65	0.10	0.19
RFN-Nest [54]	2021	29.15	1.67	6.13	6.84	0.34	1.50
CrossFuse-GAN [61]	2023	36.26	1.59	10.06	6.82	0.38	1.30
LRG-CNN [55]	2023	35.67	1.70	9.38	6.84	0.33	1.42
Ours		92.80	<i>1.23</i>	7.74	0.25	1.40	0.32

† means the larger the better. ‡ means the smaller the better.

from other methods. For example, the windows at the top left of the blue box are clear in our fused image and are not clear in the fused images from other methods. Our fused image is more pleasant to the human eyes than other fused images. Therefore, our proposed method has a strong ability to preserve salient foreground regions, objects, and details and improve the overall appearance of the fused images.

In summary, these visual comparative results illustrate good performance of our proposed method in preserving salient foreground regions, salient objects, and true input details without introducing extra artifacts. Furthermore, these visual comparative results also illustrate the excellent performance of our method in improving fused image quality, especially adaptive contrast enhancement, which leads to visual pleasant of our fused results.

D. Quantitative Results

To quantitatively compare the fusion performance of our proposed IVF method with the other 16 algorithms, comparative metrics from different methods are listed in Table II in the TNO

dataset. The best, second-best, and third-best values in the table are highlighted in red bold, green italics, and underlined blue, respectively.

As shown in Table II, our proposed method achieves five best metrics (i.e., EI, EN, MIab/f, SD, and VIFF) and two second best metrics (i.e., SF, DF) among all 17 methods including ours. The largest EI value and the second largest SF value indicate that our fused images have most details and edges. The largest EN value shows richest information in our fused images. The largest MIab/f value shows that our fused images take most information from the source input images. The largest SD value and the second largest DF values indicate that our fused images have best image contrast among all methods. The largest VIFF value means that our fused images are most natural and realistic. These observations from the quantitative metrics are consistent with our previous observations from the above-mentioned specific exemplar results in Figs. 11–14.

In summary, the proposed method in this study achieved superior performance in both quantitative and qualitative results when compared with typical methods.

V. DISCUSSION

We tried to solve the simultaneous supervised image fusion and fusion image quality enhancement problem in IVF domain in this article by providing a PGI computation method and an SSR embedded GAN network structure. The achieved experimental results are appealing.

Compared with previous GAN-based IVF methods [7], our proposed method in this article avoids the biasing and balancing problems, and also reduces the number of loss components in loss function. Our proposed general mechanism can be easily integrated into other unsupervised deep IVF methods by incorporating additional PGIs and an SSR component.

We tried the simple threshold method “Otsu”[43] to segment foreground salient regions. It is possible to improve the fusion performance by trying other advanced segmentation method. Furthermore, other advanced image quality enhancement component could also be embedded into the GAN network to further improve fusion performance.

For unsupervised IVF methods, a potential way to improve performance is to construct a loss function which is robust to poor image quality, such as poor illumination, noise, and poor contrast.

In summary, our proposed method in this study is not only able to fuse IR and VIS images, but also able to improve fused image quality with an ending to ending machine learning way.

VI. CONCLUSION

In this study, we tried to solve the shortage problem of ground truth fusion image in IVF domain by building a PGI. The PGI conception was first introduced and a simple PGI image computation method was proposed. A novel SSR embedded GAN network based on two-encoder-one-decoder was designed for efficient mixed modal feature extraction and fused image reconstruction. A dense and mixed modal inputting strategy was also introduced for better mixed modal feature extraction.

Our proposed PGI image emphasizes both IR and VIS foreground regions and salient objects. Image details are also preserved in our PGI image by our residual summation operation. The SSR component is seamlessly embedded into our dense two-encoder-one-decoder GAN structure to potentially improve fused image quality, especially the contrast of fused image. The proposed PS-GAN structure has the potential to preserve image details and our PGI image can not only help the network to preserve salient foreground regions, but also guide the network to preserve image details. Furthermore, our proposed modal mixed and dense inputting strategy also makes the mixed modal feature extraction possible with our simple dual path network.

Extensive experimental results on public IVF datasets verified the superior performance of our proposed approach over other typical methods in terms of both subjective visual quality and 11 objective metrics. It is demonstrated that both the salient foreground objects and image details between the IR and VIS images can be efficiently extracted using our proposed PS-GAN, and the fused image contrast is significantly improved. In conclusion, we offered a novel GAN mechanism to simultaneously address salient foreground region aware, detail preservation, mixed modal feature extraction, and fusion image contrast boosting problems in IVF area.

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